



IOANNIS ANGELO TASSIOULAS

Date of Birth: 21/02/2001

Education

MSc Applied Physics

September 2024 – Ongoing

Delft University of Technology

Delft, Netherlands

- Track: Physics for Quantum Devices and Quantum Computing
- Following Research and Development Track, which includes Systems Engineering course and a Professional Practice Internship
- Courses taken in first semester: Partial Differential Equations, Finite Elements, Object Oriented Scientific Programming in C++, Advanced Statistical Mechanics, Introduction to High Performance Computing, Ethics and Engineering, Fundamentals of Quantum Information
- Courses following in the second semester: Advanced Quantum Mechanics, Advanced Solid State Physics, Quantum Computer Architecture, Computational Physics, Artificial Intelligence for Physicists, Systems Engineering

BSc Physics

September 2020 – August 2024

University of Groningen

Groningen, Netherlands

- Track: Particle Physics
- Followed courses in both the elementary physics subjects such as classical mechanics, thermodynamics, electromagnetism, special relativity and also more advanced topics such as nuclear physics, quantum mechanics, group theory and quantum field theory
- Completed bachelor project on 11th of July, 2024.
- Grade point average: 7.8 over 190 ECTS

International Baccalaureate

September 2018 - May 2020

HL: Mathematics, Computer Science, Physics; SL: English L&L, French B, History

Athens, Greece

- Final Grade of 40, including a 7 in French B SL after 1 year
- HL Grades of 7, 7, 5 respectively
- Diploma given at St. Catherine's British School

Projects

Quantum Critical Behaviour in the Interacting Majorana Chain Model

July 2025-April 2026

Chepiga Group - Quantum Nanoscience, TU Delft

Delft, Netherlands

- Using a numerical algorithm, studied a particular quartic interaction in the Majorana-Kitaev Chain, a toy model for Majorana Zero Modes
- Performed preliminary literature review, built upon previously developed code, and analyzed and interpreted results.
- Project required constant clear communication with both supervisors as well as self guided motivated research.

Optimization of Sound Source Localization in Neuromorphic Systems

April 2024-July 2024

BICS - CogniGron, University of Groningen

Groningen, Netherlands

- Computational research project focused on simulating neuromorphic TDEs.
- Skills trained include project planning, scientific communication and scientific modeling via Python.
- Final outcome of project included both symposium presentation and written thesis.

Simulation of Phase Transitions in Percolating Systems

May 2025-June 2025

AP3082: Computational Physics End Project

Delft, Groningen

- Developed a computational simulation of a site percolation problem, and studied the phase transitions and physical attributes of the system via NumPy, SciPy and Matplotlib

Work Experience

Lab Inventory Management Student Assistant

June 2025 - Ongoing

QuTech

Delft, Netherlands

- Worked as a student assistant aiding in the development of a common lab inventory system for QuTech

- Engineering an ongoing software project via updating a Streamlit-based webapp and managing database entries using Excel and SQL, meeting with stakeholders and PIs to develop product and cataloguing different lab equipment and invoices
- Managed and organized meetings with financial stakeholders as well as lab visits with PIs at QuTech to fully develop final equipment catalog

Student Coach for TN1101 : Technology Management

January 2025 - June 2025

TU Delft, in collaboration with Rotterdam School of Management

Delft, Netherlands

- Working together with 1st year student in BSc Technische Natuurkunde in TU Delft, I helped foster teamwork and open communication as they studied emergent technologies for a local company in the first phase.
- In the second phase, teamed up with student coach from Rotterdam to coach nine groups of 6 students from RSM and TU Delft as they design the final report

Mathematical Physics Student Assistant

April 2025 – July 2025

FSE, University of Groningen

Groningen, Netherlands

- Tutorial TA for Mathematical Physics, course code WBPH049-05.
- Entrusted with planning and organizing tutorials, grading student assignments.
- Entrusted by professor to revise exams and proctor exam sessions.

Electronics and Signal Processing Teaching Assistant

February 2023 – March 2024

FSE, University of Groningen

Groningen, Netherlands

- Lab TA for Electronics and Signal Processing (WBPH038-05), as well as Lab and Tutorial TA for Electronics for BME (WBBE009-05).
- Experience in leading computer labs using LTSpice.
- Supervised and helped students in building physical circuits, working with opamps and operating oscilloscopes during physical labs.
- Led tutorials by solving practice problems concerning electronics and digital counters, and graded homeworks.

Extracurriculars

Buitenlandse-excursiecommissie

October 2022 - October 2023

Position: Treasurer

Groningen, Netherlands

- I took part of a committee board that organized an educational trip to the UK, visiting different physics laboratories and research departments.
- As treasurer, I organized the financing of the trip, kept a budget of all expenses and income streams, and communicated with sponsors.

StCatsHacks

September 2019 - November 2019

Organizer

Athens, Greece

- I organized the annual hackathon at St Catherine's British School in Lykovrisi, Athens.
- Along with my computer science colleagues, I helped contact sponsorships, communicate with guests from the tech industry and help plan out the scheduling and catering.

D'Arcy McGee Jazz Band/Senior Band

September 2015 - May 2018

Saxophonist

Gatineau, Quebec

- Performed as part of the D'Arcy McGee Jazz Band and Senior Concert Band, performing in both regular concerts as well as provincial competitions.
- Played alto, tenor and baritone saxophone throughout my 3 years in both bands.

Skills

Programming Languages: Python, C++, Java, SQL

Packages: SciPy, NumPy, Pandas, PyTorch, Matplotlib, Streamlit, Git, Bash

Software & Tools: Mathematica, MATLAB, LTSpice, Microsoft Office, LaTeX, Microsoft Teams

Languages: English (*Native*), Portuguese (*Fluent*), French (*Intermediate*), Spanish (*Intermediate*), Greek (*Intermediate*), Dutch (*Beginner*)